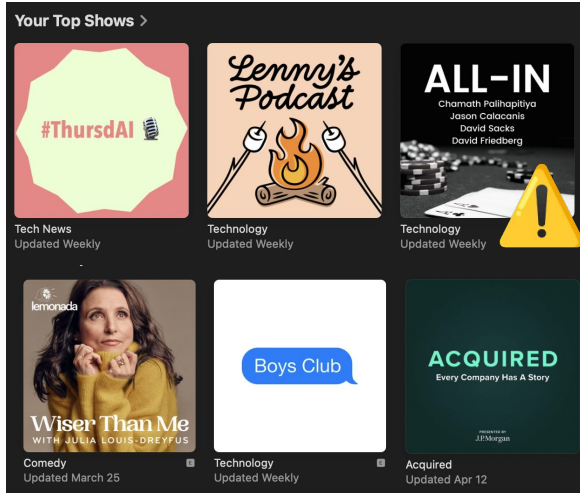


Embeddings

Module 6 Week B – Text as Vectors

Get to know Lana — Fun Fact

I love podcasts



My favorites:

- ThursdAI –Weekly AI news roundup
- Latent Space → Technical AI engineering deep dives
- Lenny's Podcast → Product and career interviews
- How I AI → Practical AI workflows (watch on YouTube)
- Wiser than Me → Iconic older women share wisdom
- Acquired → Deep dives into companies (long!)
- All-In → Four VCs on tech, markets, politics ⚠️
- EconTalk → Conversations on economics and ideas
- Boys Club → Tech, crypto, and internet culture
- HBR IdeaCast → Leadership and management insights

On my “check out sometime” list:

- 10% Happier or The Happiness Lab
- No priors → Two AI investors interviewing founders and researchers
- The Peter Attia Drive or Found My Fitness

From Entities To Texts

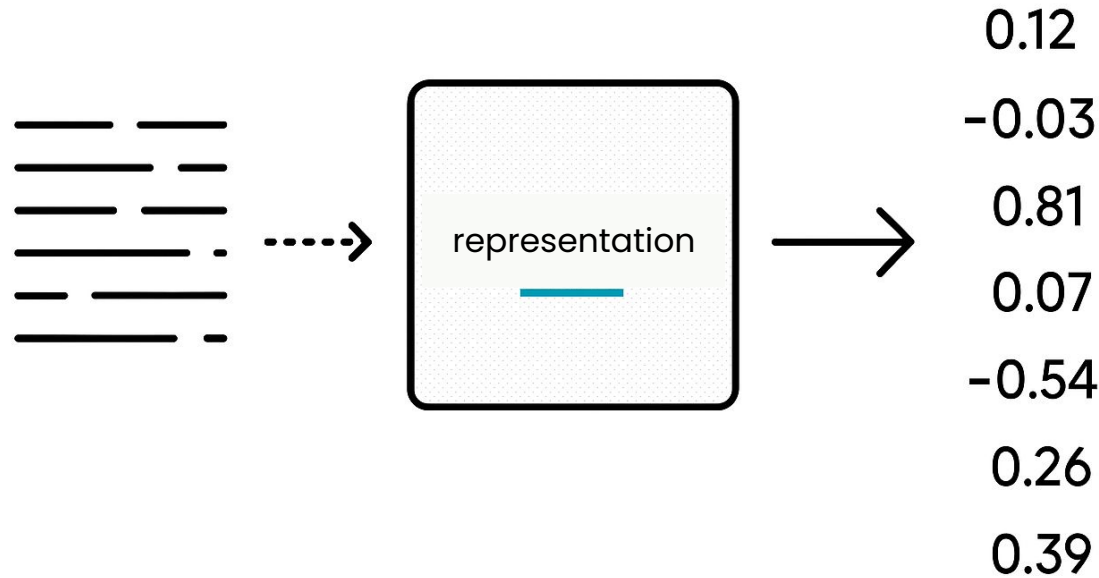
Layla launched a startup in **Amman** this week.



Which is More Similar?



Models Need Numbers



Counting Words

Bag Of Words (BoW)

VOCABULARY

climate

change

warming

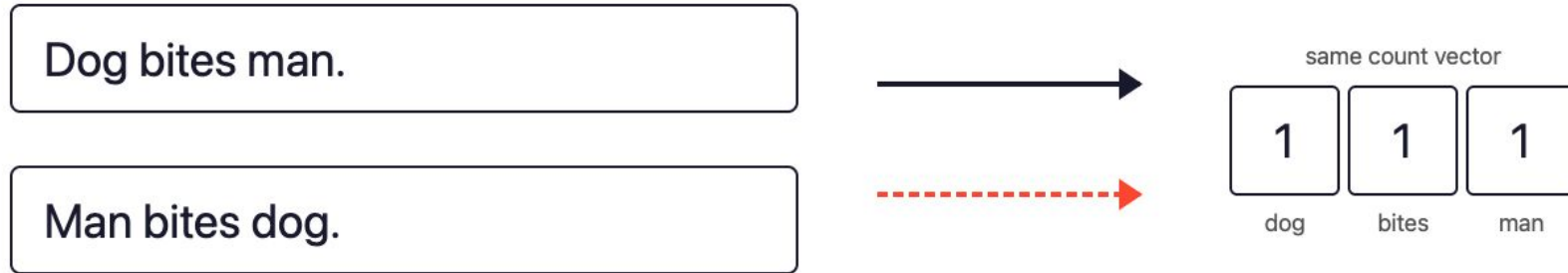
ice

fish

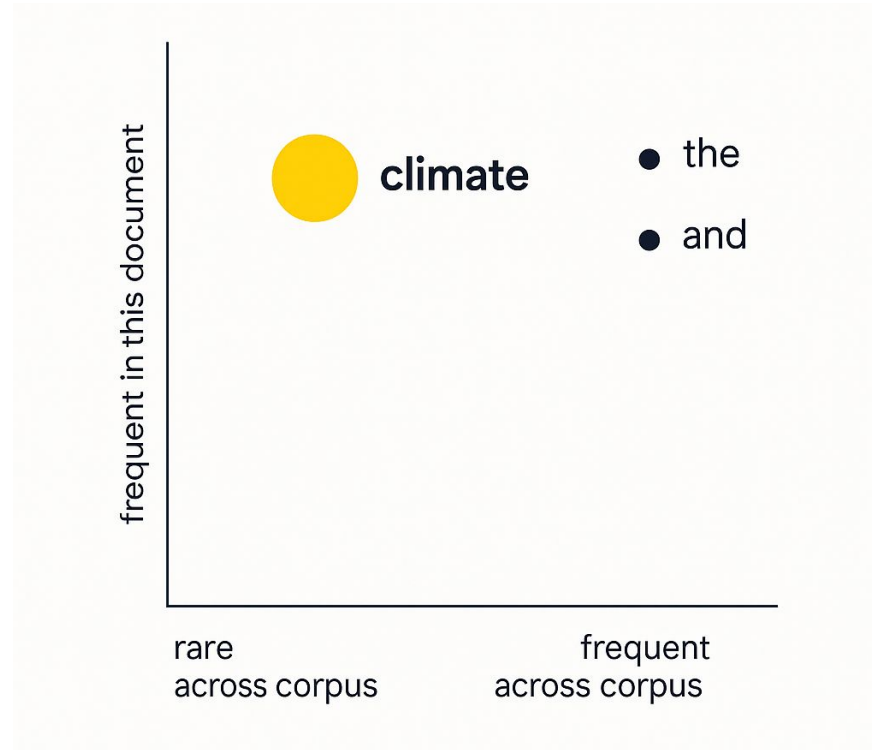
	climate	change	warming	ice	fish
Sentence 1	2	0	1	0	0
Sentence 2	1	1	0	1	0
Sentence 3	0	0	0	0	1

count vector per sentence

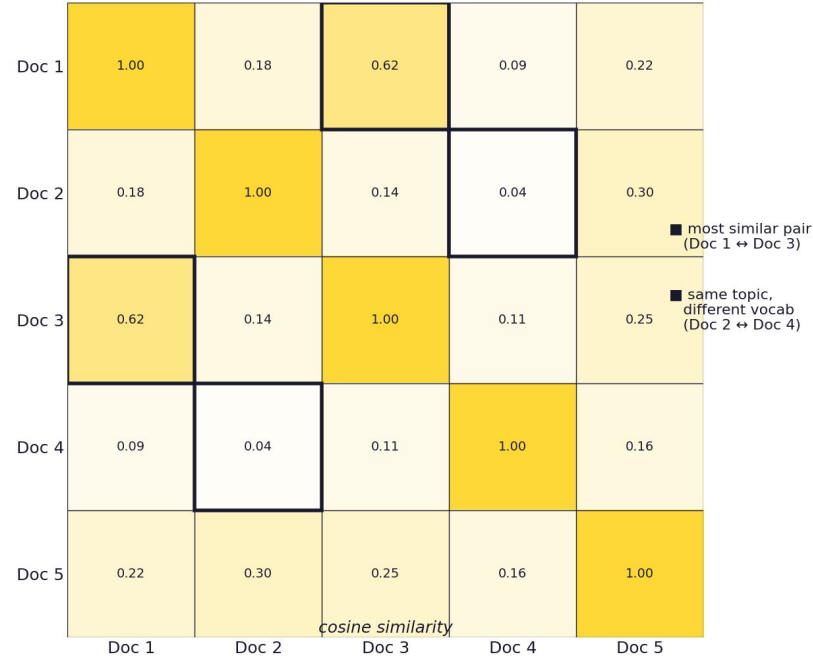
Word Order Disappears



TF-IDF



TF-IDF Similarity Matrix



When Counting Hits a Wall



Words as Vectors

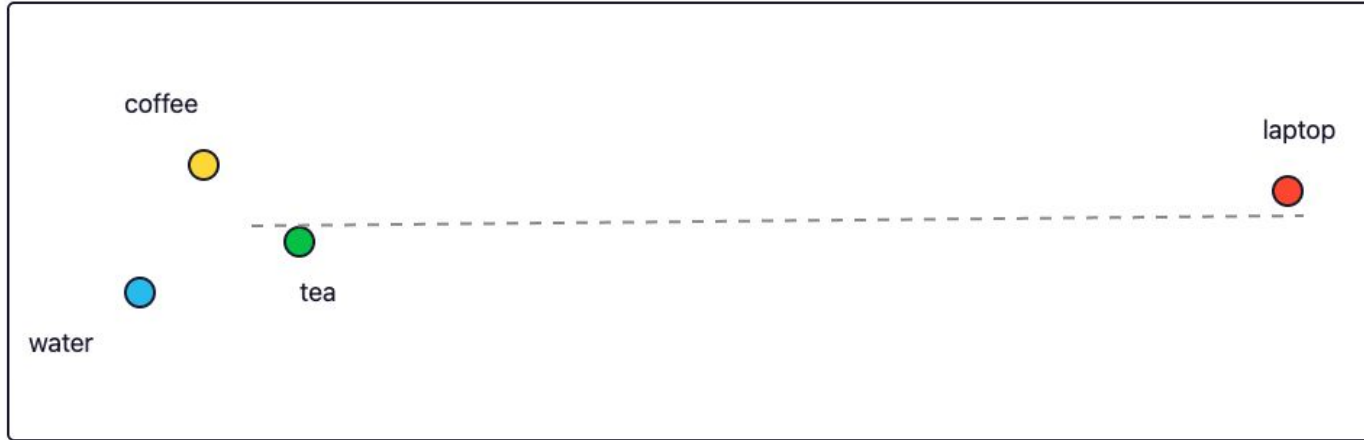
Distributional Hypothesis

I drank a cup of _____ this morning.

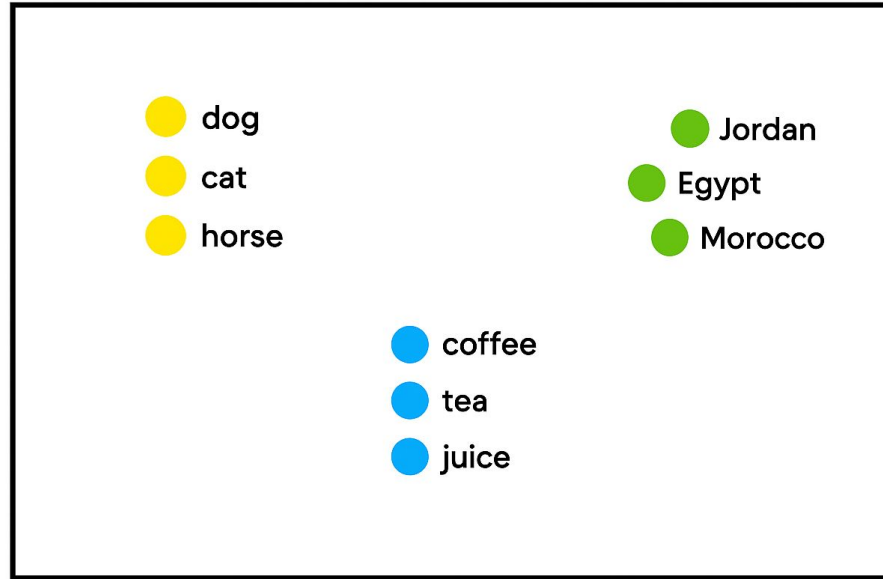
I had _____ with breakfast.

The waiter poured me a cup of _____ .

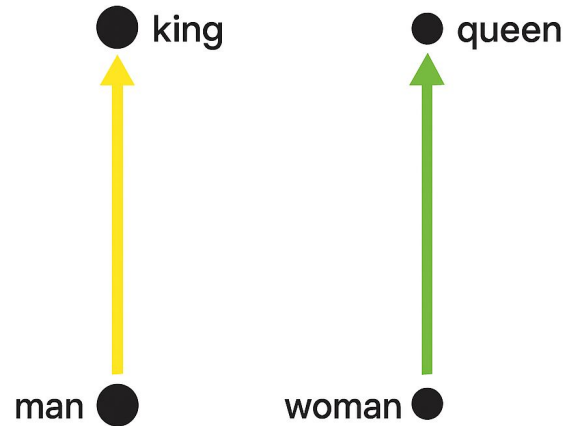
WORD VECTOR SPACE



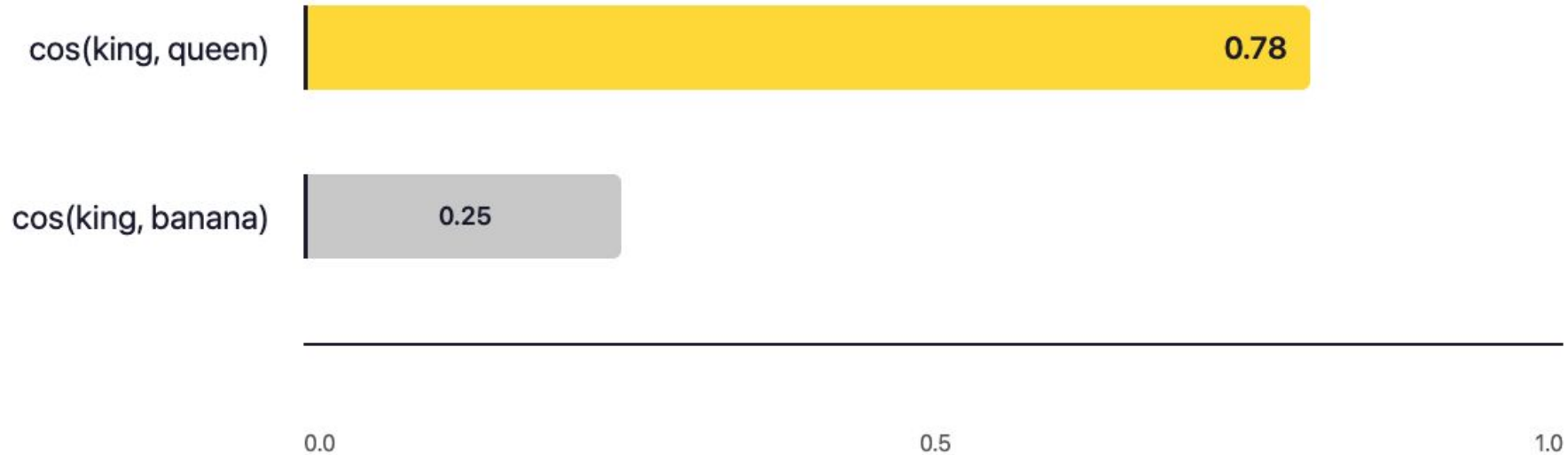
A Word In Space



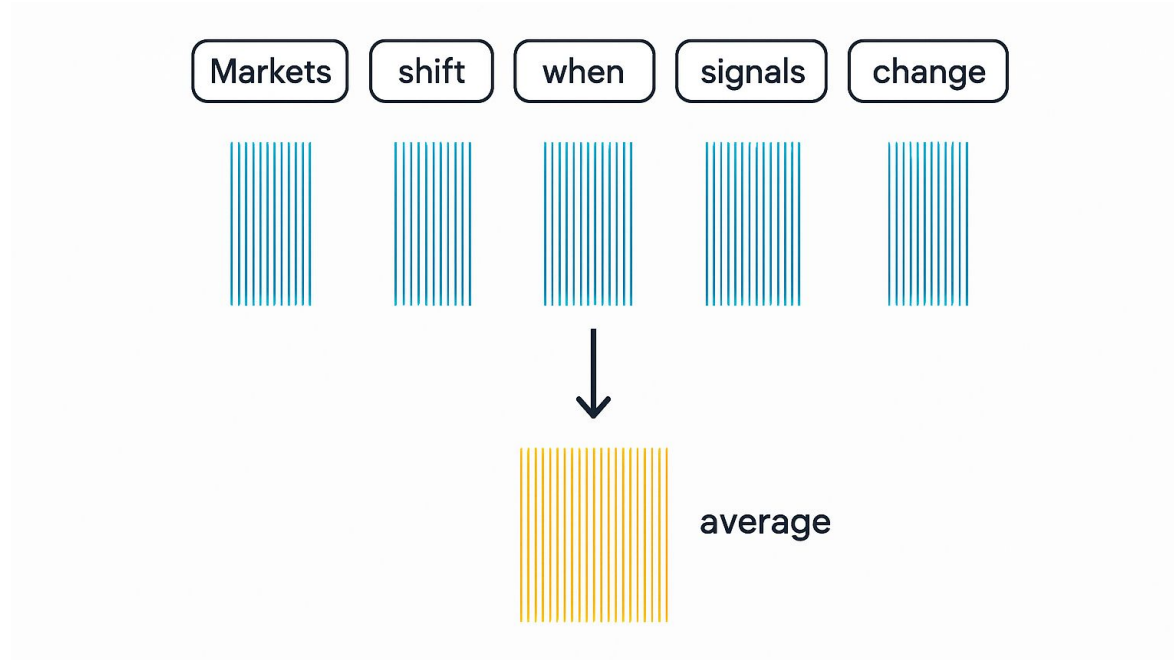
King - Man + Woman



GloVe Similarity



Document Vectors From Words



The Context Problem

The Same Word Twice

GloVe (static): one vector per word

river **bank**
bank account

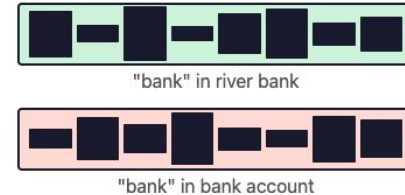
...→
 ...→



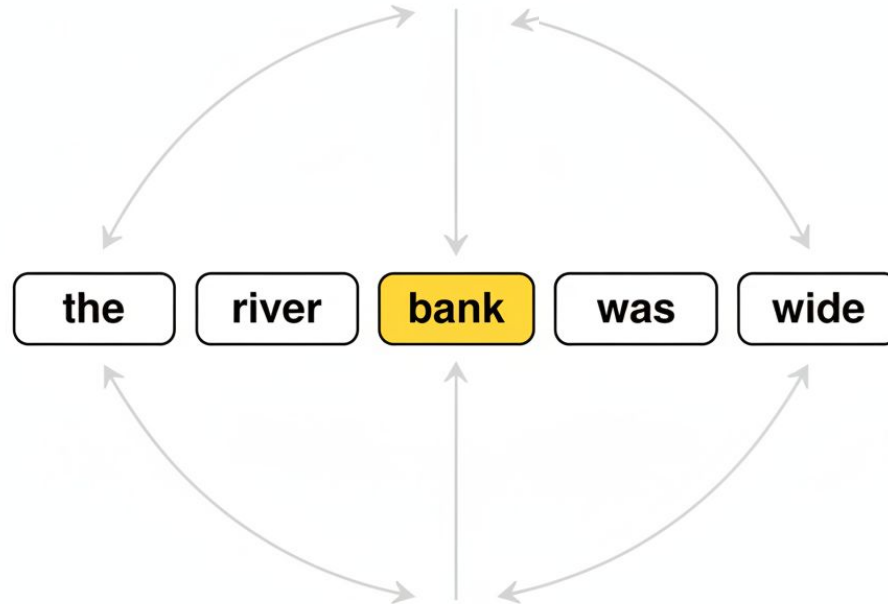
Contextual (preview): two vectors, one per meaning

river **bank**
bank account

→
 →



Every Token Looks Around



The Extraction Recipe



Two Banks, Two Vectors

I sat on the river **bank**



I opened a **bank** account



Choosing your Representation

Three Methods Side By Side

TF-IDF

A ==

B ==

C ==

GloVe

A ==

D ==

C ==

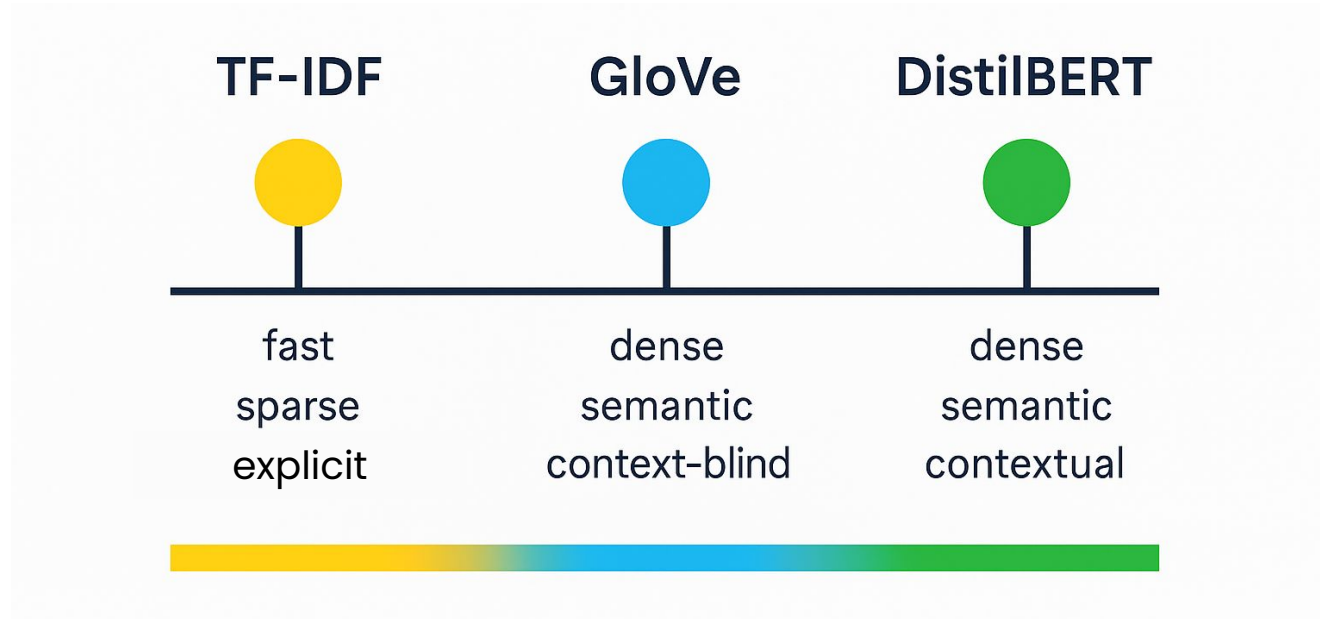
DistilBERT

A ==

D ==

E ==

The Spectrum



When To Reach For Which

Multi-step manual
approach process:

- loading tokenizer/model
- tokenizing
- forward pass
- mean pooling for each text

----- approach process:



find documents containing
these keywords

TF-IDF



group articles
by topic

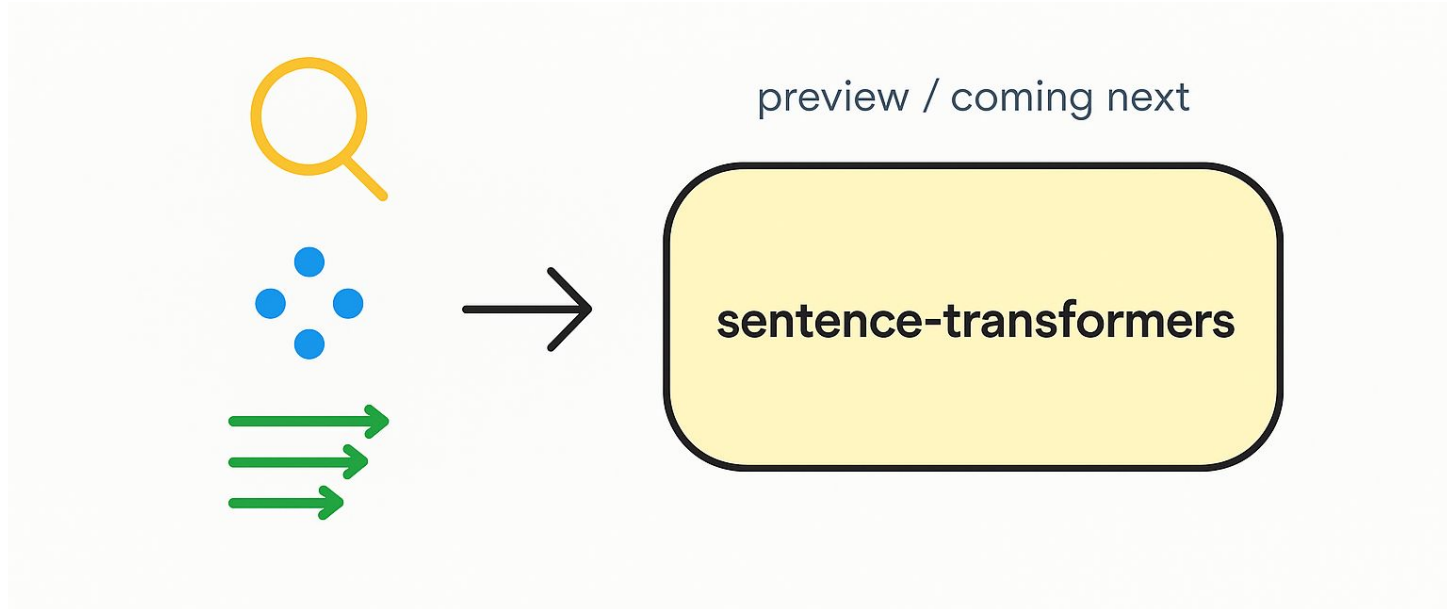
GloVe



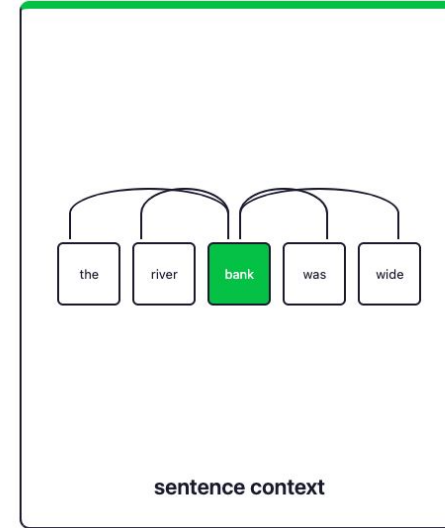
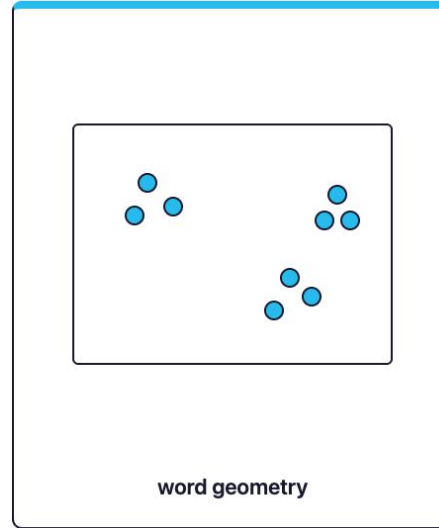
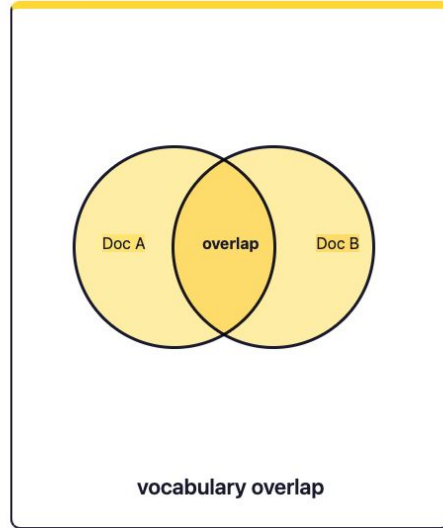
tell river bank
from bank account

DistilBERT

Sentence Transformers Preview



Three Ways To Compare



Questions?